

VibeStack

An agentic harness that turns a plain-English description into a **validated, self-healing FastAPI backend**, built with LangGraph.

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13
commits

~2,100
lines backend
(Python)

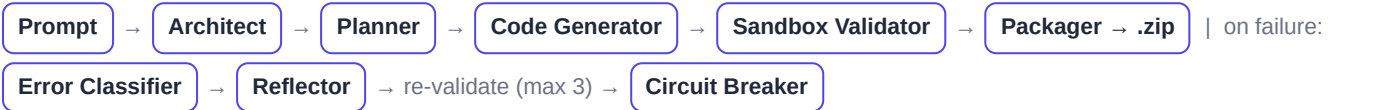
~1,500
lines frontend (TS)

35 / 35
tests passing

10
benchmark prompts

3
Canva decks

1. The product: what works today



Part	What it does	Status
Architect agent	The LLM turns the prompt into a typed data model (entities, fields, foreign keys, auth), validated with Pydantic	Done
Planner + Code Generator	Verified Jinja2 templates are the baseline; the LLM writes models, schemas and routers per entity, in parallel	Done
Sandbox Validator	3 gates: import the app → boot it and check /health → run API tests generated from the spec	Done
Self-healing loop	The error is classified (syntax, import, orm, contract...) and the faulty file located; the repair escalates cheap model → strong model → templates	Done
Circuit breaker + ledger	Stops after 3 attempts with a diagnostic report; every change is logged with a diff	Done
Live preview	The generated app runs for real and opens in the UI (sign up, create records, FK dropdowns, Swagger docs)	Done
Web UI (Next.js)	Live pipeline cards, activity feed, Architecture / Self-healing / Live App tabs, red/green fix diffs	Done
Demo bug injection	Plant a "wrong HTTP status" or "broken import" bug to show healing live	Done
Engineering	State saved at every step (SQLite checkpoints); per-run token budget and cost shown in the UI; generated code runs as an unprivileged sandbox user	Done
Deployment	Single Docker image (API + UI), docker-compose, Railway config	Done
Tests & evals	35 automated tests pass (spec, codegen, healing, graph, API). 10-prompt benchmark runner is ready	Done

2. Timeline

- 5 Oct: VibeStack MVP: LangGraph harness with self-healing
- 7 Oct: Demo flow trimmed; second bug type; live preview; richer generated apps; projector-friendly UI
- 8 Oct: First-review deck; Level 2 DFDs (processes 2.0, 3.0, 4.0); Review 1 (Interim Evaluation) deck; LangGraph seminar deck
- 9 Oct: MITS logo on every seminar slide; slide 1 guide and institute details fixed

3. Presentations & documents

First Review deck

MITS project template; monochrome diagrams with straight connectors; text-only Algorithm slide.
[Open in Canva ↗](#) · docs/presentation/VibeStack-First-Review.pptx

Review 1: Interim Evaluation

12 slides plus Thank You, following the review guidelines. Committee feedback applied: a single SDG (SDG 9), precise activities and deliverables, a percentage-only Remaining Work slide, and a plain-text timeline.
[Open in Canva ↗](#) · docs/presentation/VibeStack-Review1.pptx

Level 2 Data Flow Diagrams

Processes 2.0, 3.0 and 4.0, with processes drawn as rhombuses.
docs/presentation/dfd-level2/

Seminar: LangGraph and its Applications

20 slides; base paper Duan & Wang, arXiv:2411.18241; 6-paper literature review (all arXiv); MITS logo on every slide; guide Ms. Nimmi M K.
[Open in Canva ↗](#) · docs/seminar/

4. Still to do

Item	Notes	Status
Benchmark results with a real model	The runner exists (<code>python -m evals.run</code>), but numbers (pass rate, heal count, cost) have not been recorded in the repo yet	Pending
Human approval step	Approve the architecture before code is generated	Roadmap
Review council	Multiple agents review the generated code	Roadmap
Long-term memory, MCP tools	Reuse past fixes; connect external tools	Roadmap
Security scanning, Java output	Static analysis of generated code; a second target language	Roadmap
Seminar base-paper detail	Slides 12–15 use the paper's abstract only; upload the PDF for exact details	Optional
Security housekeeping	Revoke the OpenRouter key that was once pasted into chat; use a fresh key	Action

5. Where things are in the repo

Path	Contents
backend/agent/	The harness: graph.py (LangGraph wiring), nodes.py (agents), sandbox.py, healing.py, llm.py, prompts.py, templates/
backend/app/	FastAPI server: runs, live events (SSE), file view, download, preview proxy
backend/tests/, backend/evals/	Automated tests; 10 benchmark prompts and the runner
frontend/	Next.js user interface
docs/	PLAN.md (demo script, Q&A), review decks, DFDs, seminar deck
Dockerfile, docker- compose.yml, railway.json	Deployment

Generated from repository [nikhilj2005/major-project](#), branch `claude/loving-rubin-g77dod`, commit `07bf1cf`. Test result: 35 passed (run on 9 Oct 2026 with `FAKE_LLM=1`).